

MESH-PATTERN PAIRS 113–116 ARE NOT EQUIDISTRIBUTED WITH THE $X^{(1)}/Y^{(1)}$ FAMILY: A COUNTEREXAMPLE TO AN OPEN PROBLEM OF LV AND ZHANG

ABSTRACT. In the concluding remark of their paper on joint equidistributions of the mesh patterns 123 and 132 with minus antipodal shadings, Lv and Zhang record an unnumbered assertion, suggested by computer experiments and left there explicitly as an open problem: the four mesh-pattern pairs labelled 113–116 have the same bivariate occurrence distribution as the pairs $X_i^{(1)}$ and $Y_i^{(1)}$, $9 \leq i \leq 18$. We disprove it on $\mathfrak{S}_4$. The former pairs have no permutation with occurrence vector $(2, 0)$, whereas every pair in the latter family has exactly one. More generally, for every $n \geq 4$ the respective counts are 0 and at least $(n-1)!/6$. The failure is immediate from Lv and Zhang’s own Theorems 6.1 and 6.2, in Section 6 of the same paper, so the assertion is best read as a slip in the concluding remark rather than as a conjecture.

All pair labels follow Lv and Zhang [1]. For an ordered pair $P = (p_1, p_2)$ of mesh patterns, set

$$T_{n,k,\ell}^P = |\{\pi \in \mathfrak{S}_n : \text{occ}_{p_1}(\pi) = k, \text{occ}_{p_2}(\pi) = \ell\}|.$$

Write $c(r, s)$ for the unsigned Stirling number of the first kind. Let $\mathcal{P}$ be the four pairs labelled 113, 114, 115, 116, and let

$$\mathcal{Q} = \{X_i^{(1)} : 9 \leq i \leq 18\} \cup \{Y_i^{(1)} : 9 \leq i \leq 18\}.$$

Section 7 of [1], headed “Concluding remark”, contains the following unnumbered sentence, in which equations (1) and (2) are the two displays of Theorem 2.4 of that paper, giving respectively the joint distribution of the pairs in $\mathcal{Q}$ and its generating function: “In fact, computer experiments suggest that pairs 113 to 116 have the same joint distribution as pairs $\{X_i^{(1)}\}_{i=9}^{18}$ and $\{Y_i^{(1)}\}_{i=9}^{18}$, satisfying (1) and (2); we leave justifying this observation as an open problem.” Formalised, and weakened to the part of it that we shall use, this is the following statement. It is unnumbered in [1] and is not the only cross-pair statement made in that section, so we do not give it a name beyond its location.

Conjecture 1 ([1, Section 7], unnumbered). *For every $P \in \mathcal{P}$, $Q \in \mathcal{Q}$, $n \geq 1$, and $k, \ell \geq 0$,*

$$T_{n,k,\ell}^P = T_{n,k,\ell}^Q.$$

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Two remarks on the standing of this assertion. First, its failure is immediate from Theorems 6.1 and 6.2 of [1], proved in Section 6 of that paper, which show that each component of each of the pairs 113–116 has at most one occurrence in any permutation: the joint distribution of such a pair is supported on $\{0, 1\}^2$, while the distribution asserted for it, equation (1) of [1], is not. Second, Conjecture 4 of [1], stated in the same section, asserts of the eight pairs 119–126 exactly what the sentence above asserts of 113–116, namely that they “have the same joint distribution as $\{X_i^{(1)}\}_{i=9}^{18}$ and $\{Y_i^{(1)}\}_{i=9}^{18}$ ”. The sentence is therefore best read as a slip in the concluding remark rather than as a conjecture. What follows records the refutation and the minimal order at which it takes effect.

Theorem 2. *For every $P \in \mathcal{P}$, $Q \in \mathcal{Q}$, and $n \geq 4$,*

$$T_{n,2,0}^P = 0, \quad T_{n,2,0}^Q = (n-1)! \sum_{j=2}^{n-1} \frac{c(j-1, 2)}{j!} \geq \frac{(n-1)!}{6} > 0.$$

At $n = 4$, one has $T_{4,2,0}^Q = 1$. Consequently, Conjecture 1 is false, and $n = 4$ is the first order at which it fails.

Proof. The proofs of Theorems 6.1 and 6.2 of [1] show that each component of the pairs labelled 113 and 115 occurs at most once in any permutation. The same conclusion for 114 and 116 follows by inversion. Hence

$$T_{n,2,0}^P = 0 \quad (P \in \mathcal{P}).$$

For every $Q \in \mathcal{Q}$, Theorem 2.4, equation (1), of [1] gives, whenever $(k, \ell) \neq (0, 0)$,

$$T_{n,k,\ell}^Q = (n-1)! \binom{k+\ell}{k} \sum_{j=2}^{n-1} \frac{c(j-1, k+\ell)}{j!}.$$

Therefore

$$T_{n,2,0}^Q = (n-1)! \sum_{j=2}^{n-1} \frac{c(j-1, 2)}{j!}.$$

For $n \geq 4$, the term $j = 3$ is present, all terms are nonnegative, and $c(2, 2) = 1$. Thus

$$T_{n,2,0}^Q \geq \frac{(n-1)!}{3!}.$$

For $n = 4$, the only nonzero term is $j = 3$, since $c(1, 2) = 0$; so $T_{4,2,0}^Q = 1$.

For $n \leq 2$, no length-three pattern occurs. For $n = 3$, the only three-term subsequence is the whole permutation, so the shading is vacuous; every pair considered here has one permutation with vector $(1, 0)$, one with vector $(0, 1)$, and four with vector $(0, 0)$. Hence $n = 4$ is minimal. $\square$

Remark 3 (An explicit counterexample). In the standard notation (σ, R) for a mesh pattern, the pair labelled 113 and the pair $X_9^{(1)}$ are

$$P_{113} = ((123, R_{113}), (132, R_{113})), \quad X_9^{(1)} = ((123, R_9), (132, R_9)),$$

where

$$R_{113} = \{(0, 1), (0, 2), (1, 1), (1, 2), (2, 2), (3, 0), (3, 1), (3, 2), (3, 3)\},$$

$$R_9 = \{(0, 0), (0, 1), (0, 2), (0, 3), (1, 1), (1, 2), (3, 1), (3, 2)\}.$$

For P_{113} , the shaded rightmost column forces every occurrence to end at the final entry. The shaded cells $(a, 2)$, $0 \leq a \leq 3$, then force the middle-position entry to equal the final entry minus 1 for the 123 component, or plus 1 for the 132 component. The first entry must be the largest entry to the left of the middle-position entry that is smaller than it, and is therefore unique. Thus each component occurs at most once.

For $X_9^{(1)}$, the permutation 1234 has two occurrences of its 123 component, on positions $(1, 2, 3)$ and $(1, 2, 4)$. The unused point lies in the unshaded cell $(3, 3)$ for the first occurrence and in the unshaded cell $(2, 2)$ for the second; the other two increasing triples are excluded by the shaded cells $(1, 1)$ and $(0, 0)$. Since 1234 has no classical 132 occurrence, its occurrence vector is $(2, 0)$.

The discrepancy is not confined to the cell $(2, 0)$. On $\mathfrak{S}_4$ the joint distribution of P_{113} is

$$(0, 0) \mapsto 12, (0, 1) \mapsto 5, (1, 0) \mapsto 5, (1, 1) \mapsto 2,$$

while that of $X_9^{(1)}$ is

$$(0, 0) \mapsto 12, (0, 1) \mapsto 4, (0, 2) \mapsto 1, (1, 0) \mapsto 4, (1, 1) \mapsto 2, (2, 0) \mapsto 1,$$

so the two already differ in the cells $(0, 1)$ and $(1, 0)$ as well. The cell $(0, 0)$, on the other hand, agrees: direct enumeration gives $T_{n,0,0}^{P_{113}} = 2(n-1)!$ for $2 \leq n \leq 7$, which is exactly the value of the $k = \ell = 0$ branch of equation (1) of [1]. A computation testing only avoidance would therefore not have detected the failure.

Remark 4 (Scope). The assertion refuted here is the unnumbered cross-pair statement in Section 7 of [1]. It is distinct from the seven within-pair symmetry conjectures involving fourteen labelled patterns in Table 11, which were proved by Fang, Fu, Kitaev, and Li [2]. The other comparisons between different pairs proposed by Lv and Zhang, namely Conjectures 3 and 4 of [1] on the pairs 117, 118 and 119–126, are not considered here.

REFERENCES

- [1] S. Lv and P. B. Zhang, Joint equidistributions of mesh patterns 123 and 132 with minus antipodal shadings, *Discrete Appl. Math.* **379** (2026), 419–433, doi:10.1016/j.dam.2025.09.002.
- [2] Q. Fang, S. Fu, S. Kitaev, and H. Li, On fourteen equidistribution conjectures of Lv and Zhang and monotone mesh patterns with corner shadings, arXiv:2507.04698, 2025.